CHC6171: Principles of Secure Operating Systems

Deadline: **MAY 15th** Total Score: **100 marks**

Coursework: **Understanding and modifying an OS- File Encryption**

Mark distribution for this module is as follows; **50% Course Work + 50% Final Examination = 100%**

Learning Outcome:

(This exercise assesses LO (Learning Outcomes) 2 and 4.)

- **LO 2:** Create system-level software that modifies and extends existing operating systems. Conduct experiments designed to evaluate the performance, security, and reliability of their modifications and additions.
- **LO 4:** Demonstrate a thorough understanding of multi-threaded/process systems through the design and implementation of communicating, multi-threaded systems software.

Coursework Task Description

1. File Encryption prevents unauthorized access to files by those who physically access a computer/drive. This is especially important to protect lost/stolen devices' data/information against unauthorized access.
2. For this exercise, you will be designing software that protects data while sitting on systems' storage, and then adding and implementing this on your selected Operating System (OS), e.g., Minix.
3. Follow the steps below in order, as you might find it harder if you don't. Complete each step before moving on to the next.
    - List the essential features that a File Encryption should have. Look at similar products for this (e.g., macOS FileVault).
    - Design your software/patch to be added to your selected OS. This should involve communications with the existing parts of the OS.
    - Implement and test your code using an incremental method.
    - Run your code and perform an integration test.

Prerequisites

1. Take your time to carefully read through the coursework descriptions to understand what is expected of you. Pay attention to the specific areas that need to be addressed and consider reading the recommended Textbook for this course for more guidelines.

2. Engage in a discussion with the module leader/Teaching assistant regarding your initial thoughts on the requirements. Schedule a meeting where you can demonstrate your file encryption software to the module leader/teaching assistant so that the current state of your work can be assessed and further suggestions for improvement can be given. This initial meeting is crucial for setting the direction of your revisions.

Report Writing Guide:

Student are required to submit their coursework source codes and a report using the **attached report template**. The details of the report are as follows:

S/N	Task Description	Mark
1	A **description** of your File Encryption	5%
2	A list of functional and non-functional **requirements** and security features of File Encryption	10%
3	**Design** of your software/patch that includes communications with the OS	10%
4	**Implementation** of your File Encryption including annotated C code	15%
5	**Testing** plan for validating your software	10%
6	Description of **integrating**/adding the implemented component to OS	5%
7	**Integration testing** plan for integrating your component into the system	5%
8	Reporting the possible **limitations**, failures, and/or difficulties you experience in your work	5%
9	A **conclusion** section that includes recommendations for extending the conducted work	5%
10	**Description and reflections on improvement/revisions made.** This section is critical as it highlights the important progress made in the current version of your coursework	15%
11	**References** (IEEE referencing Style)	5%
12	Multi-thread demonstration and Seminar Class Attendance/Exercises	10%
	Total	100

Note: Through the design and implementation of your system, demonstrate a thorough understanding of multi-threaded/process systems. This is not a separate section in your report. Instead, it has to be addressed and presented in the other sections, e.g., 2, 3, 4, and 6 (5%).

50% Attendance seminar class attendance/Exercises (5%)

Submission Description:

1. Submit your final report via the student website by **MAY 15th.**
2. Your report must at most be 2000 words (excluding tables, figures, annotated codes, and references).
3. The reports longer than 10% of the word limit will be penalized; the extra words will not be marked.
4. Marks and feedback will be available on the student website as soon as all marking routines are completed.
5. This coursework is an individual piece of work. The University rules concerning plagiarism, syndication, and cheating apply.

Recommended References:

1. Textbooks:

- Operating System Concepts by Abraham Silberschatz and Peter B. Galvin John Wiley and Sons 2013
- Modern Operating Systems: Global Edition by Andrew S Tanenbaum and Herbert Bos Pearson 2014
- Operating Systems: Internals and Design Principles, Global Edition Pearson 2017

2. Internet Resources:

- File Encryption Software: The Complete Guide https://www.winzip.com/en/learn/tips/file-encryption/encryption-software/
- How to make a File Encryption Software https://www.devteam.space/blog/how-to-build-your-own-file-encryption-software/
- How to Encrypt Files to Protect Personal and Business Data? https://geekflare.com/file-encryption-software/
- What is file encryption? https://blog.box.com/what-is-file-encryption
- Encrypting a file before sharing http://kb.mit.edu/confluence/display/istcontrib/Encrypting+a+file+before+sharing
- Kaspersky- Endpoint Security for Windows 11.3.0 File Level Encryption https://support.kaspersky.com/KESWin/11.3.0/en-US/193688.htm